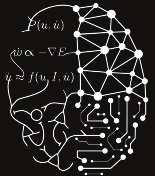
LIF and Simplified SRM Neurons Encode Signals Into Spikes via a Form of Asynchronous Pulse Sigma-Delta Modulation

Young C. Yoon [1]

GUT meeting 2026-11-10



NeuroTMA

Xavier Lesage
Review: Luke Eilers

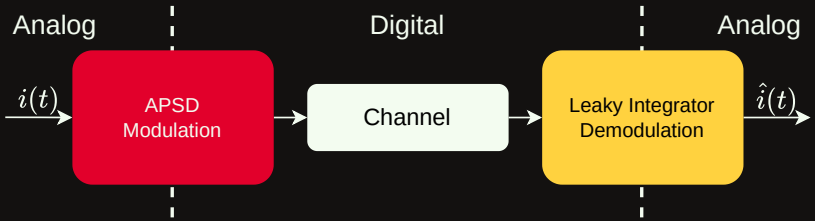
NeuroTMA, Department of Physiology, University of Bern
Tuesday 10th November, 2026

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Why spikes ?

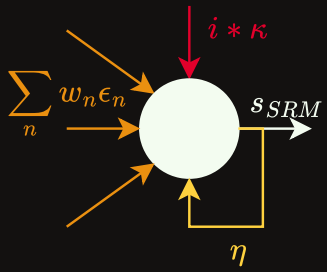
Motivation

- An approach from Analog-to-Digital conversion and digital communications
- Spiking mechanism as a conversion from analog (continuous time, continuous space) to digital (discrete time and/or discrete space)
- How can we exploit this mechanism to design digital hardware ?



Spike Response Model 0th order [2]

Spike Response Model - Neuron Model



- κ : input response
- η : output response
- ϵ : presynaptic spike response

$$v_{SRM}(t) = i(t) * \kappa(t) + \sum_{m=1}^{M_{SRM}} \eta(t - T_m^{SRM}) + \sum_{n=1}^{N} w_n \sum_{m_n=1}^{M_n} \epsilon_n(t - T_{m_n}^n)$$

$$s_{SRM}(t) \triangleq \sum_{m=1}^{M_{SRM}} \delta(t - T_m^{SRM})$$

Alternative Form

Spike Response Model - Neuron Model

$$
\begin{aligned}
y_{SRM}(t) &\triangleq i(t) * \kappa(t) \\
\hat{y}_{SRM,L}(t) &\triangleq -\sum_{m=1}^{M_{SRM}} \eta(t - T_m^{SRM}) \\
&= -\eta(t) * s_{SRM}(t) \\
v_{SRM}(t) &= y_{SRM}(t) - \hat{y}_{SRM,L}(t)
\end{aligned}
$$

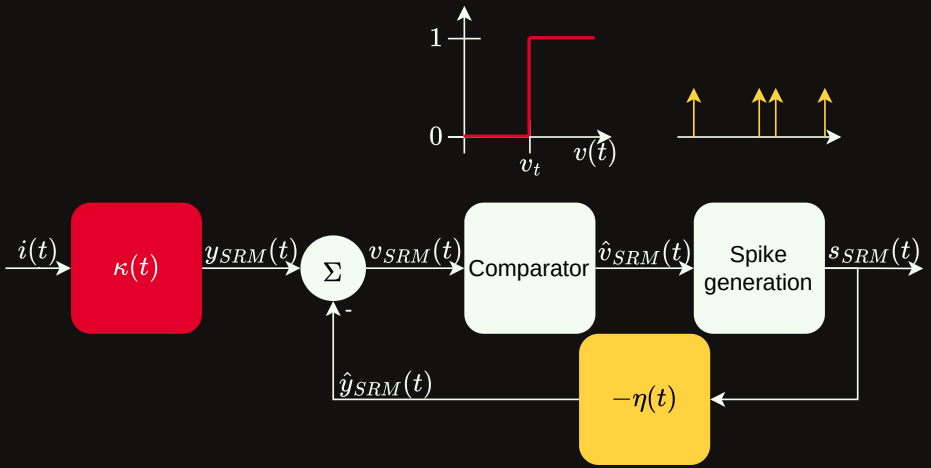
- $y_{SRM}(t)$: input current response filter
- $\hat{y}_{SRM,L}(t)$: unsmoothed reconstruction estimate of $y_{SRM}(t)$

Remark

- Tracking error is bounded by v_t

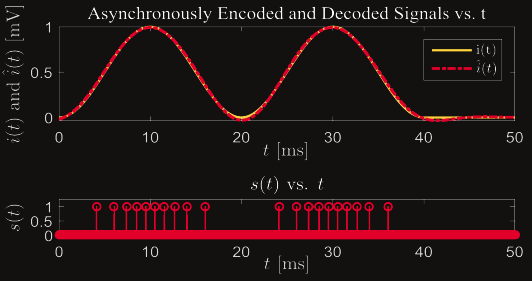
Filter Building Block

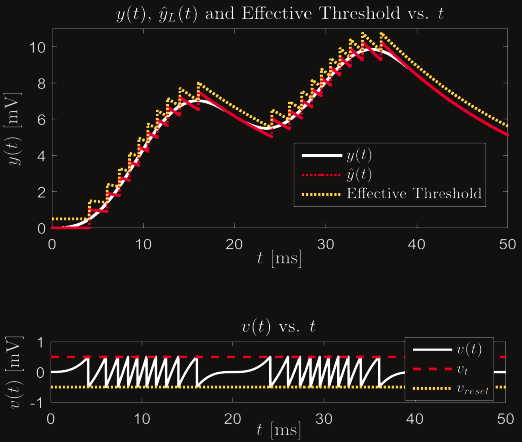
Spike Response Model - Neuron Model



Signals

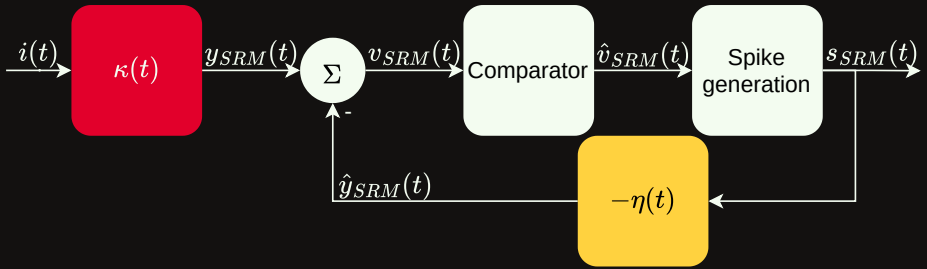
Spike Response Model - Neuron Model





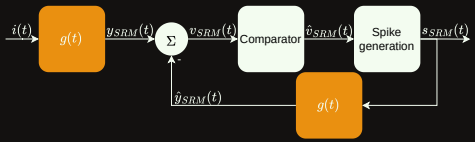
Where is the integral ?

Sigma-Delta Modulation - Canonical Form



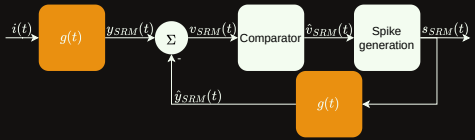
Special case

Sigma-Delta Modulation - Canonical Form

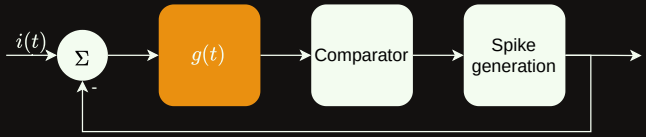


Special case

Sigma-Delta Modulation - Canonical Form

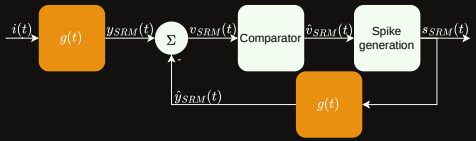


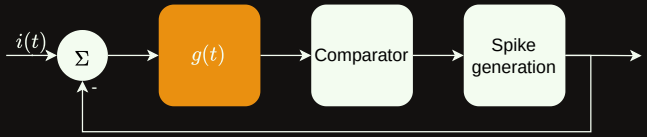
Parameter settings for equivalence of SRM_0 and APSDM Encoder	Parameter settings for equivalence of the LIF neuron and APSDM Encoder with RC Filters
$v_t = \Delta/2$	$v_t = \Delta/2$
$v_{SRM}(0) = 0$	$v_{LIF}(0) = 0$
$\kappa_0(t) = g(t)$	$h(t)/C = g_Z(t) = g(t)$
$\eta(t) = -g(t)$	$(v_t - v_{reset})h(t) = g_A(t) = g(t)$
n/a	$C = 1/(v_t - v_{reset})$



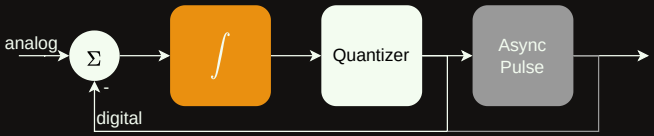
Special case

Sigma-Delta Modulation - Canonical Form

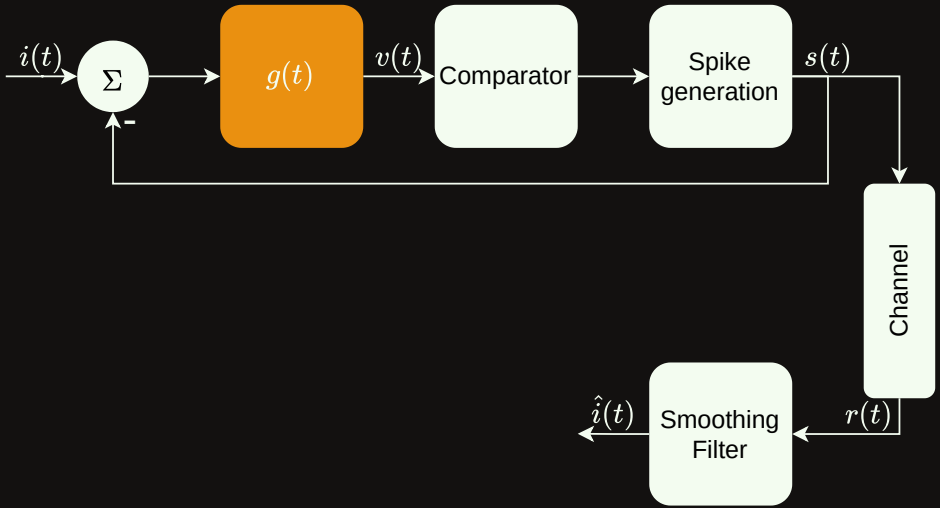




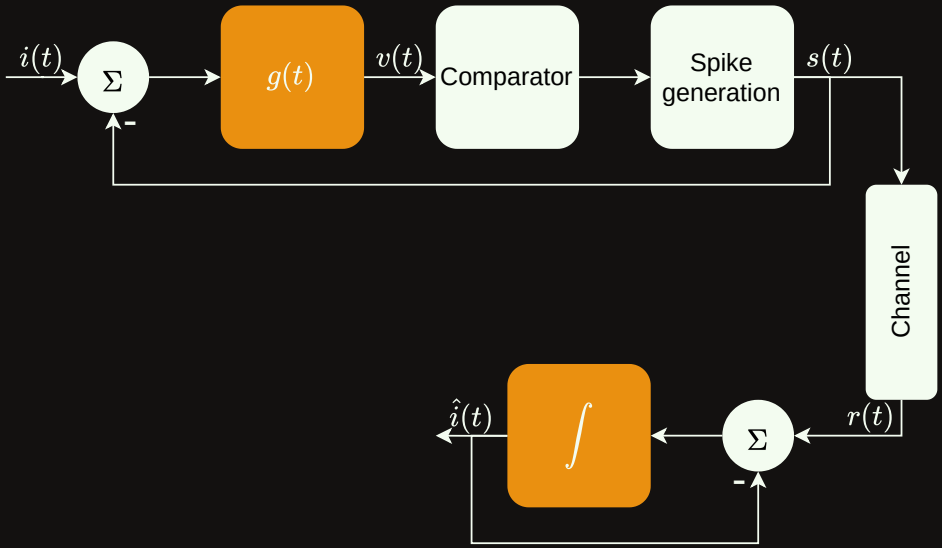
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Demodulation



Demodulation



Summary

Conclusion



- Formulation as the difference of an input response filter and its approximation
  - Approximation error and spike frequency both depend on the threshold
- Can be mapped to asynchronous pulse sigma-delta modulation
- Demodulation can be done by a low pass filter such as a Leaky Integrator

Perspectives

Conclusion

- APSDM is a simple case (unipolar, one threshold) but it can be extended (e.g. bipolar send-on-delta)
- What does it mean for processing with neurons ?
- How do we relate this with numerical solvers ?
- How can we exploit this structure to produce neuromorphic HW ?
- How biologically plausible is all of that ?
  - Neuronal Spike Generation Mechanism as an Oversampling, Noise-shaping A-to-D Converter - Chklovskii Soudry [3]

Further reading : [4], [5]

References I

[1] Y. C. Yoon, "LIF and Simplified SRM Neurons Encode Signals Into Spikes via a Form of Asynchronous Pulse Sigma-Delta Modulation," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 28, no. 5, pp. 1192–1205, May 2017, ISSN: 2162-2388. DOI: 10.1109/TNNLS.2016.2526029 Accessed: Nov. 6, 2026. [Online]. Available: https://ieeexplore.ieee.org/document/7419261

[2] W. Gerstner and W. M. Kistler, *Spiking Neuron Models: Single Neurons, Populations, Plasticity*. Cambridge University Press, 2002.

[3] D. B. Chklovskii and D. Soudry, "Neuronal spike generation mechanism as an oversampling, noise-shaping A-to-D converter," in *Proceedings of the 26th International Conference on Neural Information Processing Systems - Volume 1*, ser. NIPS'12, vol. 1, Red Hook, NY, USA: Curran Associates Inc., Dec. 2012, pp. 503–511. Accessed: Nov. 6, 2026.

[4] H. Schindler, "Linear, Nonlinear, and Adaptive Delta Modulation," *IEEE Transactions on Communications*, vol. 22, no. 11, pp. 1807–1823, Nov. 1974, ISSN: 1558-0857. DOI: 10.1109/TCOM.1974.1092119 Accessed: Nov. 6, 2026. [Online]. Available: https://ieeexplore.ieee.org/document/1092119

[5] C. Kikkert and D. Miller, "Asynchronous Delta Sigma Modulation,", vol. 36, Apr. 1975.